3D CAD Designs Documentation

By: Job Daniel Abarquez

Tree Climber

- **Name:** Tree Climber
- **Software:** Autodesk Fusion
- **Objective:** Design for coconut tree climbing prototype to reduce workload on coconut farmers during harvest season.
- **Process:** Collaboration via Discord and Excalidraw; Reference machine study; Parametric multi-component design; Final assembly
- **Status:** Concept
- **Estimated Time:** 25h 30m + rev. 10h
- **Simulation Data:** None
- **Revisions:** 1
- **Revision Notes:** Size scaled down
- **Reference:** https://doi.org/10.3390/act11100287
- **Downloadable:** Github

Tumbler

- **Name:** Tumbler
- **Software:** Autodesk Fusion
- **Objective:** Create a tumbler that has a detachable handle.
- **Process:** Sketching; initial 3D designs; parametric base; feature additions; revisions
- **Status:** Concept
- **Estimated Time:** 9h + rev. 1h
- **Simulation Data:** None
- **Revisions:** 2
- **Revision Notes:** Added strap attached on cap; Added rubber strap within a groove around cap
- **Reference:** Commercially sold tumblers
- **Downloadable:** Github

Umbrella Frame

- **Name:** Umbrella Frame
- **Software:** Autodesk Fusion
- **Objective:** Imitation design / conversion of existing object into a 3D asset.
- **Process:** Image capture; measurement; parametric base; feature completion
- **Status:** Concept
- **Estimated Time:** 10h + rev. 2h
- **Simulation Data:** None
- **Revisions:** 1
- **Revision Notes:** Added spring mechanism inside the frame
- **Reference:** Personal umbrella
- **Downloadable:** Github

Branch Trimmer Tool

- **Name:** Branch Trimmer Tool
- **Software:** Autodesk Fusion
- **Objective:** Design for functionality.
- **Process:** Sketching; initial designs; parametric base; feature addition
- **Status:** Concept
- **Estimated Time:** 1h
- **Simulation Data:** None
- **Revisions:** 0
- **Revision Notes:** No revisions
- **Reference:** Machete
- **Downloadable:** Github

Conference Table

- **Name:** Conference Table
- **Software:** Autodesk Fusion
- **Objective:** Design for functionality.
- **Process:** Sketching; initial designs; parametric base; design integration review
- **Status:** Concept
- **Estimated Time:** 2h + rev. 1h
- **Simulation Data:** None
- **Revisions:** 2
- **Revision Notes:** Make the top circular; Undo shape of the top to a rectangular shape
- **Reference:** Existing online designs
- **Downloadable:** Github

Stand Table

- **Name:** Stand Table
- **Software:** Autodesk Fusion
- **Objective:** Design for functionality.
- **Process:** Sketching; initial designs; parametric revisions
- **Status:** Concept
- **Estimated Time:** 2h + rev. 1h
- **Simulation Data:** None
- **Revisions:** 2
- **Revision Notes:** Added more extension and retraction length; Widened top and support
- **Reference:** Existing online designs
- **Downloadable:** Github

Cone Attachment

- **Name:** Cone Attachment
- **Software:** Autodesk Fusion
- **Objective:** Speed modelling.
- **Process:** 2D draft to 3D conversion
- **Status:** Concept
- **Estimated Time:** 30m
- **Simulation Data:** None
- **Revisions:** 0
- **Revision Notes:** No revisions
- **Reference:** tootalltoby
- **Downloadable:** Github

Connector

- **Name:** Connector
- **Software:** Autodesk Fusion
- **Objective:** Speed modelling.
- **Process:** 2D draft to 3D conversion
- **Status:** Concept
- **Estimated Time:** 24m
- **Simulation Data:** None
- **Revisions:** 0
- **Revision Notes:** No revisions
- **Reference:** tootalltoby
- **Downloadable:** Github

Slider Component

- **Name:** Slider Component
- **Software:** Autodesk Fusion
- **Objective:** Speed modelling.
- **Process:** 2D draft to 3D conversion
- **Status:** Concept
- **Estimated Time:** 25m
- **Simulation Data:** None
- **Revisions:** 0
- **Revision Notes:** No revisions
- **Reference:** tootalltoby
- **Downloadable:** Github

Sheet Metal Holder

- **Name:** Sheet Metal Holder
- **Software:** Autodesk Fusion
- **Objective:** Speed modelling.
- **Process:** 2D draft to 3D conversion
- **Status:** Concept
- **Estimated Time:** 25m
- **Simulation Data:** None
- **Revisions:** 0
- **Revision Notes:** No revisions
- **Reference:** tootalltoby
- **Downloadable:** Github

Slab Comp

- **Name:** Slab Comp
- **Software:** Autodesk Fusion
- **Objective:** Speed modelling.
- **Process:** 2D draft to 3D conversion
- **Status:** Concept
- **Estimated Time:** 30m
- **Simulation Data:** None
- **Revisions:** 0
- **Revision Notes:** No revisions
- **Reference:** tootalltoby
- **Downloadable:** Github

Hinge Component

- **Name:** Hinge Component
- **Software:** Autodesk Fusion
- **Objective:** Speed modelling.
- **Process:** 2D draft to 3D conversion
- **Status:** Concept
- **Estimated Time:** 35m
- **Simulation Data:** None
- **Revisions:** 0
- **Revision Notes:** No revisions
- **Reference:** tootalltoby
- **Downloadable:** Github

Angled Holder

- **Name:** Angled Holder
- **Software:** Autodesk Fusion
- **Objective:** Speed modelling.
- **Process:** 2D draft to 3D conversion
- **Status:** Concept
- **Estimated Time:** 30m
- **Simulation Data:** None
- **Revisions:** 0
- **Revision Notes:** No revisions
- **Reference:** tootalltoby
- **Downloadable:** Github

Static Angle Component

- **Name:** Static Angle Component
- **Software:** Autodesk Fusion
- **Objective:** Speed modelling.
- **Process:** 2D draft to 3D conversion
- **Status:** Concept
- **Estimated Time:** 45m
- **Simulation Data:** None
- **Revisions:** 0
- **Revision Notes:** No revisions
- **Reference:** tootalltoby
- **Downloadable:** Github

Tubes and Shelves

- **Name:** Tubes and Shelves
- **Software:** Autodesk Fusion
- **Objective:** Speed modelling.
- **Process:** 2D draft to 3D conversion
- **Status:** Concept
- **Estimated Time:** 35m
- **Simulation Data:** None
- **Revisions:** 0
- **Revision Notes:** No revisions
- **Reference:** tootalltoby
- **Downloadable:** Github

Soap Dish

- **Name:** Soap Dish
- **Software:** Autodesk Fusion
- **Objective:** Speed modelling.
- **Process:** 2D draft to 3D conversion
- **Status:** Concept
- **Estimated Time:** 25m
- **Simulation Data:** None
- **Revisions:** 0
- **Revision Notes:** No revisions
- **Reference:** tootalltoby
- **Downloadable:** Github

Bearing Connector

- **Name:** Bearing Connector
- **Software:** Autodesk Fusion
- **Objective:** Speed modelling.
- **Process:** 2D draft to 3D conversion
- **Status:** Concept
- **Estimated Time:** 25m
- **Simulation Data:** None
- **Revisions:** 0
- **Revision Notes:** No revisions
- **Reference:** tootalltoby
- **Downloadable:** Github

Sheet Metal Holder 2

- **Name:** Sheet Metal Holder 2
- **Software:** Autodesk Fusion
- **Objective:** Speed modelling.
- **Process:** 2D draft to 3D conversion
- **Status:** Concept
- **Estimated Time:** 30m
- **Simulation Data:** None
- **Revisions:** 0
- **Revision Notes:** No revisions
- **Reference:** tootalltoby
- **Downloadable:** Github

Oxygen Tank

- **Name:** Oxygen Tank
- **Software:** Autodesk Fusion
- **Objective:** Speed modelling.
- **Process:** Image reference; parametric base; add small features
- **Status:** Concept
- **Estimated Time:** 1h 15m
- **Simulation Data:** None
- **Revisions:** 0
- **Revision Notes:** No revisions
- **Reference:** Laboratory Equipment
- **Downloadable:** Github

Angle Grinder

- **Name:** Angle Grinder
- **Software:** Autodesk Fusion
- **Objective:** Speed modelling.
- **Process:** Image reference; parametric base; add small features
- **Status:** Concept
- **Estimated Time:** 3h 15m + rev. 10m
- **Simulation Data:** None
- **Revisions:** 1
- **Revision Notes:** Added power cable
- **Reference:** Laboratory Equipment
- **Downloadable:** Github

One Second Make

- **Name:** One Second Make
- **Software:** Autodesk Fusion
- **Objective:** Speed modelling.
- **Process:** Apply scripting; parametric base; add features
- **Status:** Concept
- **Estimated Time:** 30m
- **Simulation Data:** None
- **Revisions:** 0
- **Revision Notes:** No revisions
- **Reference:** Laboratory Equipment
- **Downloadable:** Github

Slider Component 2

- **Name:** Slider Component 2
- **Software:** Autodesk Fusion
- **Objective:** Speed modelling.
- **Process:** 2D draft to 3D conversion
- **Status:** Concept
- **Estimated Time:** 30m
- **Simulation Data:** None
- **Revisions:** 0
- **Revision Notes:** No revisions
- **Reference:** tootalltoby
- **Downloadable:** Github; convert to MSword-ready